

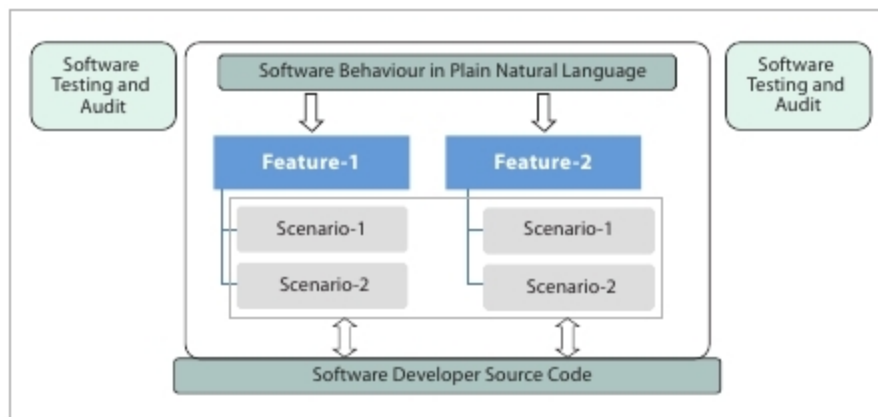


Figure 1: Key elements of behaviour driven development (BDD)

BDD describes the detailed view and process by which the end user makes use of a particular software application. In addition, the BDD approach enables a higher level of communication and collaboration among development team members. This approach is more diversified and focused towards the ability of the system to present a detailed view.

In the case of BDD, the story framework is implemented to describe the detailed architecture and flow of the process to be followed.

In the story framework, there are three key components.

- **Role:** User ('As a')
- **Feature:** Requirements ('I want')
- **Benefit:** Outcome ('So that')

In the story framework, the scenarios are associated so that the internal conditions of the test process can be integrated.

Open source frameworks for behaviour driven development and DevOps

Framework	Program- ming language	URL
Behat	PHP	http://behat.org/en/latest/
Behave	Python	https://behave.readthedocs.io/
Concordion	Java	https://concordion.org/
Cucumber	Ruby	https://cucumber.io
Cucumber-js	JavaScript	https://github.com/cucumber/cucumber-js
DaSpec	JavaScript	https://github.com/daspec/daspec-js
JBehave	Java	https://jbehave.org/
JBehave Web	Java	https://github.com/jbehave/jbehave-web
Jasmine	JavaScript	https://jasmine.github.io/
Kahlan	PHP	https://github.com/kahlan/kahlan
Lettuce	Python	http://lettuce.it/

Behat: A PHP based framework for BDD

Behat (<http://behat.org>) is a powerful, multi-featured framework with a behaviour-driven development (BDD)

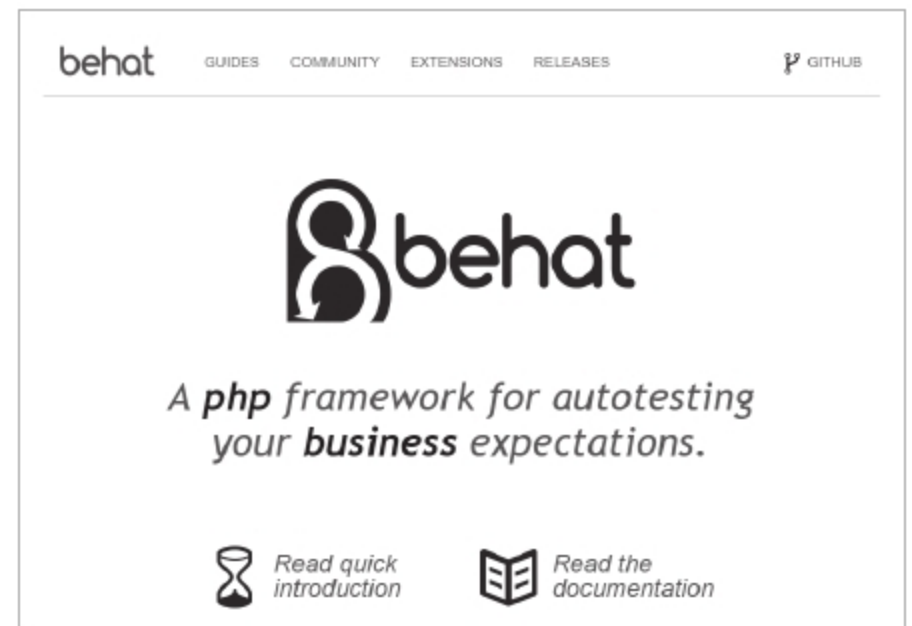


Figure 2: Official Web portal of Behat

strategy. It is developed in PHP, and supports the syntax structure of traditional sentences in the English language so that the process of test case writing is easy and understandable to all.

Behat is distributed as free and open source with many functionalities and flexibilities, enabling users to customise it for a higher degree of accuracy and performance in the test cases.

Installation of the Behat framework

According to the official documentation, to work with the Behat framework, the minimum version of PHP is 5.3.3.

Installation Method 1 — Using Composer: Using Composer is the official method to install Behat. Composer refers to the package manager for PHP. In addition to the fresh installation of Behat, the new released version can be updated using this approach.

Once Composer is integrated with PHP installation, the following instruction is executed in the project directory:

```
$ php composer.phar require --dev behat/behat
```

After successfully running the above mentioned instruction, the version of Behat that's been installed can be displayed using the following command:

```
$ /bin/behat -V
```

Installation Method 2 -- Using PHAR: This is another method to install Behat. In this case, download the *behat.phar* file from <https://github.com/Behat/Behat/releases>. After downloading, the *behat.phar* file is moved to the project directory, and then the following command can be used to check the installed version of Behat:

```
$ php behat.phar -V
```

Here, the *.phar* file refers to the PHP archive. It is a